

THAARU Sepsis AI — Temporal Sequence Report

Sequence Generation Summary • Generated: July 28, 2026

1. Executive Summary

This phase generated **1,330,534** temporal sequences across **5** experiment configurations. Each sequence is a fixed-length window of standardized clinical features, paired with a binary sepsis label aligned to the specified prediction horizon.

2. Experiment Dataset Summary

Config	Window	Horizon	Train Seqs	Val Seqs	Test Seqs	Train +Rate
w6_h0	6h	+0h	250,711	54,473	53,052	2.23%
w12_h0	12h	+0h	206,426	44,982	43,558	2.29%
w24_h0	24h	+0h	125,533	27,602	26,181	2.91%
w12_h3	12h	+3h	184,765	40,335	38,918	2.37%
w12_h6	12h	+6h	163,786	35,808	34,404	2.50%

3. Validation Results

w6_h0: ■ ALL PASSED

w12_h0: ■ ALL PASSED

w24_h0: ■ ALL PASSED

w12_h3: ■ ALL PASSED

w12_h6: ■ ALL PASSED

4. Per-Configuration Details

w6_h0 — Minimal observation: can 6h detect sepsis?

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	250,711	5,596	245,115	2.23%	43.8:1
Validation	54,473	1,224	53,249	2.25%	43.5:1
Test	53,052	1,211	51,841	2.28%	42.8:1

w12_h0 — Standard 12h observation window

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	206,426	4,720	201,706	2.29%	42.7:1
Validation	44,982	1,080	43,902	2.40%	40.6:1
Test	43,558	1,009	42,549	2.32%	42.2:1

w24_h0 — Extended observation: does more history help?

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	125,533	3,657	121,876	2.91%	33.3:1
Validation	27,602	871	26,731	3.16%	30.7:1
Test	26,181	739	25,442	2.82%	34.4:1

w12_h3 — Early warning: 3-hour advance prediction

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	184,765	4,373	180,392	2.37%	41.2:1
Validation	40,335	1,005	39,330	2.49%	39.1:1
Test	38,918	924	37,994	2.37%	41.1:1

w12_h6 — Early warning: 6-hour advance prediction

Split	Sequences	Positive	Negative	+Rate	Imbalance
Train	163,786	4,091	159,695	2.50%	39.0:1
Validation	35,808	946	34,862	2.64%	36.9:1
Test	34,404	857	33,547	2.49%	39.1:1